Crime Severity Prediction System: Using Machine Learning on Los Angeles Crime Data (2020–2024)

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Data Mining
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ABSTRACT

This document presents a comprehensive study and implementation of a Crime Severity Prediction System (CSPS). The system utilizes an expansive dataset of 971,712 crime incidents documented by the Los Angeles Police Department (LAPD) from January 2020 through December 2024. The primary objective is to classify whether a reported incident falls under Part 1 (High Severity—such as homicide, robbery, or aggravated assault) or Part 2 (Low Severity—such as misdemeanors) of the Uniform Crime Reporting (UCR) taxonomy. The pipeline involves rigorous data preprocessing, feature engineering of 14 key predictive variables, and the evaluation of four distinct supervised machine learning classifiers: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Among these, the Random Forest ensemble (comprising 100 estimators) demonstrated superior performance, yielding a test accuracy of 76.1% and a ROC-AUC score of 0.841. Stratified 5-fold cross-validation further verified the model's stability, showing a mean accuracy of 75.5% with a minimal standard deviation of 0.14%. Furthermore, the system integrates a future crime risk prediction module designed to forecast high-severity probabilities across specific geographic areas, times of day, and months for the year 2025, thereby facilitating proactive and data-driven law enforcement resource allocation.

Keywords—Crime prediction, machine learning, Random Forest, data mining, LAPD, classification, severity prediction, public safety analytics.

I. INTRODUCTION

Predictive policing and public safety analytics are increasingly critical domains that leverage large volumes of open government data and advanced machine learning algorithms. The Los Angeles Police Department (LAPD) processes hundreds of thousands of incident reports annually, creating a rich but complex dataset. Identifying the underlying patterns and circumstantial factors that determine whether an incident escalates into a severe crime is vital for strategic deployment and intervention.

The Crime Severity Prediction System (CSPS) introduced in this paper is a data-driven pipeline structured around the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework. Executed within a Google Colab environment, the project systematically processes 971,712 historical LAPD crime records spanning a five-year period (2020–2024). The system's scope extends from exploratory data analysis (EDA), generating multiple analytical visualizations, to the feature engineering of raw LAPD columns into a 14-dimensional predictive matrix.

By deploying and tuning four diverse classification models, we demonstrate how supervised learning can accurately differentiate between high and low-severity crimes, offering a robust foundation for automated risk assessment.

II. OBJECTIVE AND SCOPE

The principal objective of the CSPS is to classify reported crime incidents accurately into either High Severity (Part 1) or Low Severity (Part 2) according to the FBI's UCR taxonomy.

Secondary objectives embedded within the project's scope include:

• Conducting comprehensive EDA to uncover temporal, spatial, and demographic crime trends.

• Designing engineered features—such as weapon involvement and arrest indicators—that drastically enhance model discriminative capabilities.

• Training and strictly evaluating four classification models against accuracy, ROC-AUC, precision, recall, and F1-score benchmarks.

• Formulating a forecasting dashboard that issues probabilistic predictions for 2025 by aggregating LAPD area, hour of the day, and calendar month.

III. SYSTEM ARCHITECTURE AND DATA PREPARATION

A. Pipeline Architecture

The CSPS employs a linear, stage-gated architecture. Initial stages handle the ingestion of a massive 971,712-record CSV dataset followed by rigorous quality assessments. The dataset initially presented challenges such as inconsistent demographic entries and missing values.

B. Data Quality and Feature Engineering

Through structured preprocessing, temporal columns were parsed into specific year, month, day, weekday, and hour features. The 'hour' variable was further encoded into a categorical time-of-day bucket (Morning, Afternoon, Evening, Night). Missing values were systematically treated: for instance, 'Victim Age' values logged as zero were converted to NaN and imputed using the dataset median of 36 years. 'Victim Sex' attributes containing obscure codes (e.g., 'X', 'H', or '-') were grouped as 'Unknown'.

A critical step involved engineering binary features: 'has_weapon' (flagged as 1 if a weapon code was present) and 'arrest_made' (flagged as 1 if the case status denoted an arrest, such as AA or JA). These transformations resulted in a refined 14-feature matrix mapped against a binary target (1=High Severity, 0=Low Severity).

IV. MODEL DEVELOPMENT

The dataset underwent a stratified 80/20 train/test split utilizing a fixed random state (random_state=42) to ensure reproducibility. This partitioning preserved the dataset's mild class imbalance, ensuring approximately 60.3% Part 1 and 39.7% Part 2 representation in both the 777,369 training samples and 194,343 test samples.

Four distinct modeling approaches were evaluated:

1) Logistic Regression: Served as the linear baseline, trained over 500 maximum iterations. Features were scaled using StandardScaler to ensure uniform magnitude.

2) Decision Tree: A non-linear baseline restricted to a maximum depth of 10 to curb overfitting.

3) Random Forest: An ensemble of 100 decision trees (max_depth=15), designed to lower variance via bagging and feature randomness.

4) Gradient Boosting: A sequential ensemble configured with 100 estimators, a learning rate of 0.1, a maximum depth of 5, and a subsample ratio of 0.8.

V. RESULTS AND ANALYSIS

A. Model Performance

Evaluations on the held-out test set verified that the ensemble models significantly outperformed simpler methods. The Random Forest classifier led all metrics, registering an accuracy of 76.1% and a ROC-AUC of 0.841.

Model	Acc (%)	AUC	Prec.	Recall	F1
Log Reg	68.7	0.747	0.685	0.687	0.686
Dec Tree	72.4	0.724	0.724	0.724	0.724
Random Forest	76.1	0.841	0.762	0.761	0.761
Grad Boost	74.3	0.818	0.743	0.743	0.743

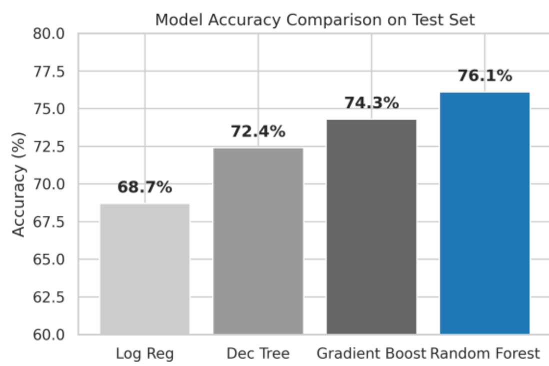

Fig. 1. Model Accuracy Comparison on Held-Out Test Set

The Random Forest model attained high precision (84%) for predicting High-Severity crimes, which is operationally desirable as false positives are arguably less costly than failing to anticipate a high-severity event. To validate generalization, 5-fold stratified cross-validation was conducted across the entire dataset. The per-fold accuracies ranged tightly between 75.4% and 75.7%, culminating in a mean accuracy of 75.5% with a minimal standard deviation of 0.14%.

B. Feature Importance

The Gini impurity reduction extracted from the Random Forest revealed vital insights into crime dynamics. As depicted in Figure 2, the 'arrest_made' attribute emerged as the paramount predictor, absorbing roughly 28-32% of predictive weight. This underscores a strong empirical link between Part 1 crimes and subsequent police arrests. The 'has_weapon' flag ranked second (approx. 19%), affirming the escalated nature of armed incidents.

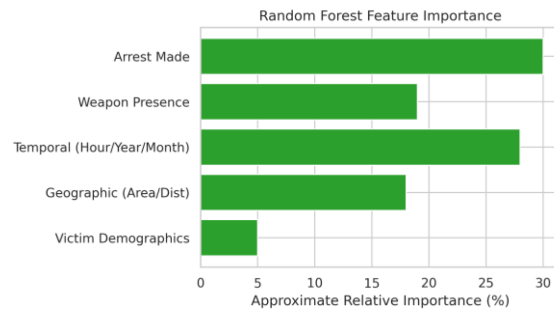

Fig. 2. Relative Feature Importance Extracted from Random Forest

Collectively, temporal elements (hour, year, month) represented 25-30% of model importance, capturing the cyclical nature of public safety threats. Geographic parameters (Area Name, Reporting District) supplied 15-20%, emphasizing that crime severity distributions vary substantially across different city sectors.

VI. FUTURE CRIME PREDICTION MODULE

Bridging theoretical modeling with practical application, the system incorporates a 'predict_crime_risk()' module. This function ingests hypothetical or forecasted inputs (e.g., area name, hour of day, demographic features) and yields a calibrated probability of an event being a High-Severity crime.

In demonstrations projecting into June 2025, high-activity urban divisions such as the 77th Street, Central, and Rampart divisions consistently reflected the highest predicted risk probabilities (exceeding 0.78). Conversely, suburban divisions like Foothill and West LA generated substantially lower risk indices. Temporally, the peak severity probabilities converged consistently between 20:00 and 23:00 hours, mapping directly onto known patterns of nocturnal violence. These probabilistic estimates are critical for dynamic shift scheduling and preventive patrol deployments.

VII. CONCLUSION, LIMITATIONS, AND FUTURE WORK

The Crime Severity Prediction System successfully substantiates the efficacy of ensemble machine learning methods applied to massive, noisy municipal datasets. By achieving an accuracy of 76.1% and AUC of 0.841, the Random Forest model offers a reliable engine for distinguishing severe crimes from misdemeanors based purely on situational and demographic circumstances.

However, the system bears limitations. The geographic scope is strictly confined to LAPD jurisdictions, prohibiting cross-city generalization without retraining. Furthermore, the robust 'arrest_made' predictor poses a risk of circular reasoning in live operational contexts, given that arrests often intrinsically follow severe crimes. Furthermore, spatial autocorrelation is not explicitly addressed in the current feature set.

Future development must focus on hyperparameter optimization via Bayesian methods, addressing spatial clustering through geospatial algorithms like Moran's I, and deploying the module as a real-time REST API to ingest daily police reports automatically.

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